

**Causal Inference 2026**  
**Assignment 2**  
**Analysis of Observational Studies**

Due: Wednesday, May 27th 2026

**Guidelines**

This assignment requires the use of R. Do not include code in your written report: instead, provide a clear and concise presentation of your results, avoiding copy-pasted R output.

The report must not exceed **5000 characters (including spaces)**, excluding tables and figures. All tables and figures must be numbered and placed at the end of the report.

The report may be written in Italian or English and must be submitted in PDF format. Partial credit based on your code may be awarded for some questions.

- Upload both your report and R code file on Moodle before the deadline.
- Use the following file naming convention:

ASS2\_GroupX.pdf      and      ASS2RCode\_GroupX.R,

where  $X$  is the group number (see the google file)

- Include the name and surname of all group members in both the PDF report and the R script.

**Data Description**

The dataset `Karolinska.txt` (available on Moodle) contains data on an observational study from the Karolinska Institute in Stockholm, Sweden. Interest focuses on the treatment of cardia cancer patients in Central and Northern Sweden, and whether it is better for these patients to be treated in a large or small volume hospital, where volume is defined by the number of patients with that type of cancer treated in recent years. The data set has 158 cardia cancer patients diagnosed between 1988 and 1995, 79 diagnosed at large volume hospitals.

The dataset includes the following variables:

- Covariate: **age** - patient's age at diagnosis; **rural** - rural area (0=urban, 1=rural); **male** - patient's gender (0=female, 1=male)
- Treatment variable: **hvdiag** (0= patient was diagnosed at a small volume hospital, 1=patient was diagnosed at a large volume hospital);
- Outcome variables: **survival**: Indicator for whether a patient survived longer than 1 year after the diagnosis.

## Questions

1. ***Assess balance and overlap in the covariates:*** Briefly comment the results.
  - (a) Inspect the number of units, and the number of units assigned to each treatment group. Summarize the data with descriptive statistics, showing the mean for all units and within each treatment group
  - (b) For each covariate, calculate the normalized difference,  $\Delta$ . For the continuous covariate `age`, also compute the dispersion balance measure  $\Gamma$ . Compare the distributions of each covariate between treatment groups using graphical methods.
2. **Stratification based on the discrete covariates.** Categorize `age` as follows:  $[25 - 60)$ ,  $(60, 75]$ ,  $> 75$ .
  - (a) Construct strata based on the following discrete covariates: categorized `age`, `rural` and `male`. Inspect the number of units, and the number of units assigned to each treatment group in each age group.
  - (b) How many strata have only treated or control units? What is the proportion of the units in these strata? How many strata have not only one treated unit or only one control unit? What is the proportion of the units in these strata?
  - (c) Drop strata where there are only treated or control units and strata where there is only one treated unit or only one control unit. Then perform a stratified analysis of the observational study. Report the point estimator, variance estimator, and 95% confidence interval for the average causal effect.
3. **Outcome regression.** Apply Lin (2013)'s estimator of the average causal effect. Report the coefficient of the treatment and the robust standard error.
4. **Estimate of the propensity score.** Estimate the propensity score for each unit in the observational study using the fitted values from a logistic regression with main effects for all of the covariates contained in the data set.
  - (a) Discard control units with estimated propensity scores lower than the minimum of the active treated units' estimated propensity scores or higher than the maximum of the active treated units' estimated propensity scores. How many units did you discard? What are we doing, and why is it important to discard these units?
  - (b) Re-estimate propensity scores on the new sample.
  - (c) Compare the distributions of the estimated propensity scores for the two treatment groups: Show summary statistics, histograms and balance measures. Comment on what you see.
5. Use the units remaining after trimming and the propensity score estimated after trimming. Select one of the following adjustment method and answer the corresponding question.

***Propensity score sub-classification.***

- (a) Create five subclasses based on the estimated propensity score using the following quantiles of the estimated propensity score: 0.25, 0.50, 0.70, 0.85. Create a table showing the number of treated and control units within each of the five subclasses.
- (b) Investigate balance in the covariate distribution before and after propensity score sub-classification, showing the Normalised differences before and after subclassification and plot them using the love-plot. Assess balance in the covariates using each covariate as an outcome.
- (c) Estimate the average causal effect, estimate the standard error (conditioning on the estimated propensity score) and report the 95% confidence interval of the average causal effect
- (d) Derive the Lin's estimator further adjusting for `age`

***Propensity score weighting.*** Considering the weights based on the estimated propensity scores and the Hájek approach.

- (a) Investigate balance in the covariate distribution before and after propensity score weighting, showing the Normalised differences before and after propensity score weighting and plot them using the love-plot. Assess balance in the covariates using each covariate as an outcome.
- (b) Estimate the average causal effect using the Hájek estimator, and estimate the standard error using bootstrap
- (c) Estimate the average causal effect using the double robust estimator, and estimate the standard error using bootstrap

**6. Matching.** Use a matching method to estimate the average treatment effect on the treated population

- (a) Create a match subsample using 1-to-1 matching with replacement with exact matching on `male` and `rural`
- (b) Investigate balance in the covariate distribution before and after matching
- (c) Estimate the average causal effect on the treated using the Biased-adjusted estimator with bias adjustments with respect to `age`. Report, its sample variance, and its 95% confidence interval.